

Jessica Snyder

Mechanical Engineering and
Mechanics Department,
Drexel University,
3141 Chestnut Street,
Philadelphia, PA 19104
e-mail: jes592@drexel.edu

Ae Rin Son

Mechanical Engineering and
Mechanics Department,
Drexel University,
3141 Chestnut Street,
Philadelphia, PA 19104
e-mail: as3266@drexel.edu

Qudus Hamid

Mechanical Engineering and
Mechanics Department,
Drexel University,
3141 Chestnut Street,
Philadelphia, PA 19104
e-mail: qh25@drexel.edu

Wei Sun¹

Mechanical Engineering and
Mechanics Department,
Drexel University,
3141 Chestnut Street,
Philadelphia, PA 19104;
Department of Mechanical Engineering,
Tsinghua University,
Beijing, China
e-mail: sunwei@drexel.edu

Fabrication of Microfluidic Manifold by Precision Extrusion Deposition and Replica Molding for Cell-Laden Device

A PED (precision extrusion deposition)/replica molding process enables scaffold guided tissue engineering of a heterocellular microfluidic device. We investigate two types of cell-laden devices: the first with a 3D microfluidic manifold fully embedded in a PDMS (polydimethylsiloxane) substrate and the second a channel network on the surface of the PDMS substrate for cell printing directly into device channels. Fully embedded networks are leak-resistant with simplified construction methods. Channels exposed to the surface are used as mold to hold bioprinted cell-laden matrix for controlled cell placement throughout the network from inlet to outlet. The result is a 3D cell-laden microfluidic device with improved leak-resistance (up to 2.0 mL/min), pervasive diffusion and control of internal architecture. [DOI: 10.1115/1.4031551]

Keywords: PED, replica molding, cell-laden microfluidic device, multinozzle cell printing

1 Introduction

Cell-laden microfluidic devices recreate in vivo cell-to-cell interactions, a current challenge in drug screening, through reproducible quantifiable patterning of spatial and temporal gradients [1–3]. The current drug discovery process leverages monolayer petri-dish model methodology to characterize pharmaceutical's absorption, distribution, metabolism, excretion, and toxicity profiles with human cells [4]. However, the fidelity of the monolayer model is limited by atypical, nonanatomical geometric simplicity, and unforced convection of mass species causes local depletion of drug-metabolizing enzymes and transporters [5]. Failure to predict drug toxicity by standard petri-dish methodology accounts for 90% of new drugs failures during Phase 1 preclinical tests [6]. Building a four-dimensional spatial and transient biomimetic environment with mathematically defined, anatomically relevant cell-to-cell contact and controllable gradients of soluble cues enables the next generation of physiologically based pharmacokinetic models for high throughput measurement of toxicity [7–9].

Advantages of 3D cell-laden microfluidic devices include controlled convection and mass transfer due to laminar flow patterns, key spatial and hemodynamic design facets of 3D topology from in vivo niche microenvironments such as the manifolds of open lumen architecture, biomimetic interstitial fluid volume to cell-laden matrix ratio of approximately one, and uniformity of

temperature field and gas supply due to the microlength scales [10–13]. Without embedded microfluidic networks for perfusion, 3D cell-laden models are limited to less than 200 μm from a surface exposed to nutrient bath to the cell core [14]. Else, the cell-laden construct's viability becomes diminished, necrotic regions develop, and phenotypes lose stability. With microfluidic techniques that are able to embed pervasive diffusion, cultures stability, consequently in vitro fidelity, increases. Cell-to-cell communication via cytokines, paracrine signaling, and soluble cues is also diffused along greater distances and between heterocellular regions, similar to vascularized in vivo tissues.

State-of-the-art cell-laden microfluidic devices have successfully leveraged embedded, directional diffusion networks to study cytokines and growth behavior mechanisms in vasculature and metastatic prostate cancer [15,16], the apoptosis-inducing effect of cytokine releasing stem cells located upstream of a cancer model [17], and 3D capillary morphogenesis and drug toxicity in hepatocytes with nonparenchymal and stem cell co-culture [18,19].

Two current challenges yet to be fully addressed by leading cell-laden microfluidic devices are 3D cell models, and consequently, 3D diffusive networks [20,21]. Enabling technology for scale-up of built biological models beyond 2D planar culture surfaces must consider additive biomanufacturing techniques to build volumes laden with cells, matrix, and pore space [22,23]. Methodological advancement of cell-laden microfluidic device fidelity relies on the microfabrication technology to control both the network's pattern of channels (curvature, bifurcations, and combined flow) and internal features (channel cross section) [24,25]. Engineering the porosity of network decreases resistance

¹Corresponding author.

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of mass transport by shortening diffusion pathways improves permeability [26,27]. Controllable scaffold parameters are pore size and an interconnected pore structure, which optimizes phenotype specific surface area preferences, cell migration, and mass transport profiles of nutrients and waste [28–31]. In vitro models balance biomimic authenticity with biomanufacturing limitations to improve model fidelity [32].

Key technology for methodological progress in lab-on-a-chip bioanalytic microsystems includes microfabrication techniques and 3D cell patterning [33,34]. Transient biological phenomenon, such as cell phenotype stability, differentiation and taxis, emerges from the key architectural design facets captured in scaffold design [35,36]. Process control of advanced biomanufacturing enables methodological progress of scaffold innovation and optimization to guide cells toward genomic stability, intercellular communication, and eventually tissue-level function [37,38]. Additive manufacturing of internal topologies enhances microstructural feature control of anatomical arrangements to build then grow mature, functional tissue from the bottom-up over time [39,40]. Direct-write bioprinting emerged as a powerful tool to directly place cells, matrix, and pore space in a 3D volume as an initial condition; limitations include loss of shape due to low mechanical rigidity and low feature resolution (50 μm) does not allow for embedded diffusion networks [41]. Fuse deposition modeling (FDM) addresses the current limitation of bioprinting with high resolution and porosity [42,43].

Replica molding further builds on the strengths of FDM to miniaturize manifolds for high-throughput microfluidic device with fine control of internal microstructures [44,45]. Other candidate techniques to build a template for replica molding include stereolithography or selective laser sintering (SLS). The high resolution of SLS successfully fabricates anatomically accurate models from medical imaging for tissue engineering, both to decrease surgery time and fabricate implants [46]. Freeze-casting template structure embedded in ceramic slurries successfully generates porosity on nano-, micro-, and mesoscale [47]. However, the process control, and consequently reproducibility and optimization, of freeze-cast pore placement is less than direct extrusion or sintering techniques. 3D printing automates geometric design freedom along a continuum of compliant to rigid forms within a single built construct [48].

The familiarity and accessibility of the FDM lower the initial cost and training time, resulting in greater traction as method. The

accessibility of FDM enables flexibility to couple an FDM templated microfluidic substrate with other advanced manufacturing methods, such as direct-write bioprinting. A combined FDM, replica molding and direct cell bioprinting process, generates a mechanical rigidity built biological system with an embedded pervasive, perfusion. Controlled cell assembly is required to produce specific cellular microenvironments and heterogeneous cell-to-cell contact [49]. Stable structural forms, as either the walls of a channel or the surfaces of a lattice structure, establish the cell densities and intercellular species transfer between cells embedded in a filament or individually encapsulated in microdroplets [50]. Stability and coordinated communication driven by physical proximity is necessary for tissue level function of a cell aggregate over time.

The objective of the presented work is to define a method to design and fabricate a microfluidic manifold, and then leverage the manifold as either (1) a mold to directly print cells into the open microfluidic channels or (2) use the structure of the interconnected 3D manifold to support and guide cell growth. The microfluidic manifold fabrication offers computer aided design and manufacturing control of the macroscale (10^{-3} to 10^{-1} m) flow pattern from inlet to outlet including internal features and microscale (10^{-4} to 10^{-3} m) channel cross section to accommodate cell and fluid volume. The authors present the set of process, design, material, and geometric parameters to define the process' capability, an empirical model to predict the flow rate to fabricate the microscale cross section of the channel is derived, maximum flow rate, and images of cellular arrangement in the microfluidic device.

2 Materials and Method

2.1 Process Overview. We investigate two types of cell-laden devices: the first with a 3D channel networks fully embedded in a PDMS substrate and the second a channel network on the surface of the PDMS substrate for cell printing directly into device channels. Fully embedded networks are leak-resistant with simplified construction methods. However, surface channel networks are partially exposed, which allows for controlled cell seeding by multinozzle deposition throughout the network. Heterocellular arrangement throughout the microfluidic device improves the in vitro model's biomimetic attributes and holds promise to increase in vitro model fidelity. Figure 1 shows two

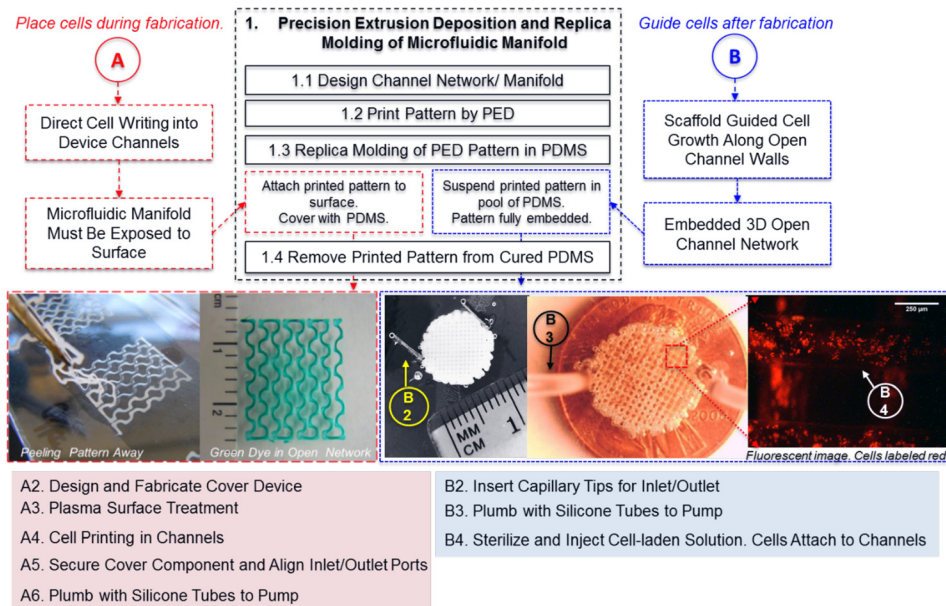


Fig. 1 Cell-laden microfluidic device construction process overview to leverage PED and replica molding to both place cells during fabrication or guide cells after fabrication

processes to either place cells during fabrication or guide cells after fabrication which both leverage a microfluidic manifold fabrication process using the FDM technique PED to print a sacrificial template followed by replica molding.

2.2 Microfluidic Manifold Fabrication Process. The microfluidic manifold fabrication process utilizes solid freeform fabrication of a sacrificial template by PED and replica molding. Step 1 is the design of the macroscale flow pattern of the manifold (10^{-3} to 10^{-5} m) and microscale (10^{-4} to 10^{-4} m) channel cross section. Step 2 is printing the design by PED. Steps 3 and 4 are replica molding and removal of the printed pattern, respectively.

Step 1 is the design of the microfluidic network to capture the key facets of mature, functional tissue within both manufacturing limitations and in vivo microfluidic networks. The characterization of the presented method's resolution and results are essential for the next iteration of in vitro model design.

Step 2 is printing a polycaprolactone (PCL), $(C_6H_{10}O_2)_n$ pellets (Sigma-Aldrich) sacrificial template by the extrusion-based solid freeform fabrication technique PED. PED has previously demonstrated the process control of internal features [51] with applications to scaffold guided tissue engineering, both in vitro and in vivo mouse [52]. Figure 2 shows a photograph of the PED system with annotations, a schematic of screw extruder with printed filament and time-lapse photographs of a single-layer scaffold printed. PCL material selection is guided by material process-ability, not biological relevance, as the PED printed pattern is a sacrificial template. The set of design variables are predicated on the tool path and the extruded filament's cross section. Separate control of the extruder printhead's height above the printing substrate and the deposition rate on the substrate diminishes the co-dependency of the cross section's height and width. The PED's process information pipeline utilizes three sub-systems: (1) the design model inputs to the automation software; (2) the motion system generates a series of two-dimensional tool paths, stacked layer-by-layer to produce a 3D volumetric construct; and (3) the material delivery system to liquefy and extrude controlled quantities of thermoplastic material.

Step 3 is replica molding of the printed pattern in PDMS, selected for biocompatible, optically transparent, and highly gas permeable properties [53]. Sylgard 184 Silicone Elastomer (Dow Corning) base and curing agent mixed 10:1, respectively, by weight is either (1) placed on a glass or polystyrene dish and covered with freshly mixed PDMS or (2) placed on freshly cured PDMS and covered with freshly mixed PDMS.

Step 4 is removal of the printed pattern from the fully cured PDMS. Printed patterns attached to a substrate and covered with

PDMS are manually removed using forceps. The embedded printed pattern is removed from fully cured PDMS by alternating thermal and chemical treatment destructive to the PCL but not PDMS. The PDMS surface is punctured by two capillary tips, one tip is an inlet plumbed to a programmable syringe tip using silicone tubing and the other is a drain. Once heated to $65-70^\circ\text{C}$, PCL melts and perfusion from the syringe pump begins 0.2 mL/hr and increased to 1.0 mL/min. An alternating cycle is to dissolve the PCL by sonication with dichloromethane at room temperature for 30–60 min.

Figure 3 shows a block diagram of the thermal/chemical cycle to remove PCL from fully cured PDMS. An ejection of water and PCL from the microfluidic manifold is identified by a black arrow. Fully cleared microfluidic manifolds are perfused by variable color dye to demonstrate the clear network of channels from inlet to outlet.

2.2.1 PED Process Controllability. The PED system extrudes molten PCL through a restrictor die along a 3D tool path. The set of design variables, Ψ , is a function of the geometric parameters, X , process parameters, Ω , and working material's properties, Φ , Eq. (1) and listed in Table 1. Process parameters are automated and can be adjusted dynamically during the build cycle period to fabricate gradient structures and multiple design variables in a single printed construct. As such, the process parameters are a function of time, t

$$\Psi(t) = f\{X, \Omega(t), \Phi\} \quad (1)$$

PED controllable process parameters are screw speed, N , temperature of material delivery system, T_i , extruder head speed, v , and height of the restrictor die outlet above the printing substrate, z , Eq. (2). The temperature of the extruder barrel is regulated by two independent heating elements: the first near the material inlet to melt the polymer and the second near the outlet. The extruder printhead speed and height are automated and their values can be programmed to dynamically change during a build cycle. Such variables are presented as a function of time

$$\Omega(t) = [N(t) \quad T_i \quad v(t) \quad z(t)] \quad (2)$$

Geometric parameters are measured using photographs of the as-built system and the image processing software IMAGEJ (NIH). Internal features of the extruder die could not be photographed and were defined using the manufacturer provided engineering drawing of the extruder die. The set of PED geometric parameters are all specific to the barrel, screw, and die of the extruder and include barrel diameter, D , restrictor die diameter, d , screw channel depth, H , pumping zone length, L , die length, l , screw channel width, W , die

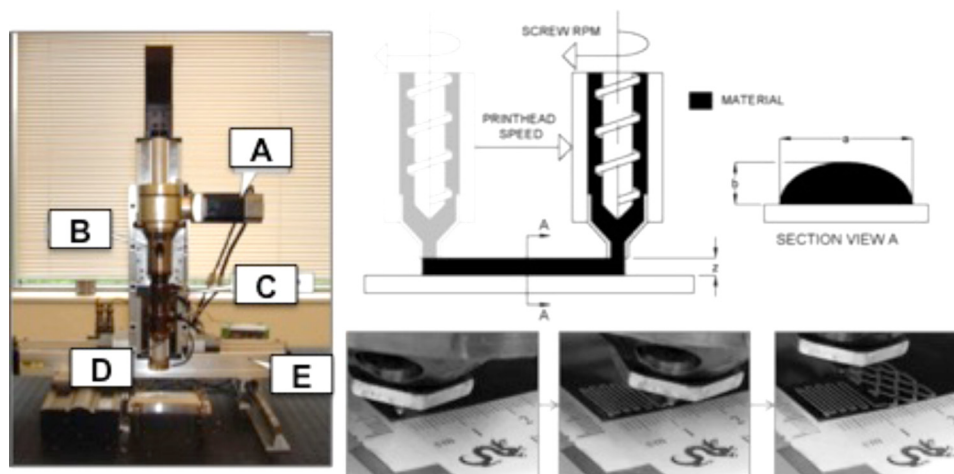


Fig. 2 Photograph of PED system with annotations (A) screw motor, (B) material inlet, (C) heating element, (D) nozzle, and (E) motion system. Schematic of PED extruder screw with process parameters and extruded filament geometrically annotated. Time-lapse photograph of single-layer scaffold printing.

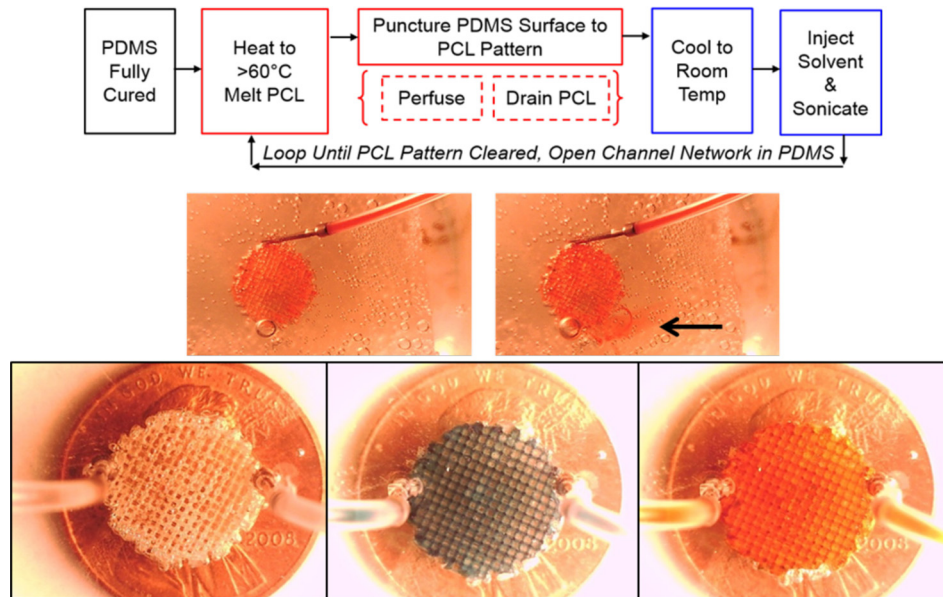


Fig. 3 Block diagram of the thermal/chemical cycle to remove PCL pattern from PDMS substrate. The black arrow points to ejection of water and molten PCL. Photographs of the fully cleared manifold perfused by variable dye fluids.

Table 1 PED system design variables, derived, geometric, material, and process parameters

Symbol	Description	Process window	Parameter type
a	Printed filament width	—	Design
b	Printed filament height	—	Design
D	PED extruder barrel diameter	20.0 mm	Geometric
d	PED extruder die diameter	100, 150, 200, 250, 300, 350, 500 μm	Geometric
H	PED extruder screw channel depth	3.0 mm	Geometric
K	PCL viscosity coefficient	$2.0 \times 10^7 \text{ Pa}$	Material
L	PED extruder pumping zone length	150 mm	Geometric
l	PED extruder die length	4.57 mm	Geometric
N	PED screw RPM	$0.1 \leq N \leq 100.0 \text{ rpm}$	Process
n	PCL viscosity index	-2.46	Material
Q	Net volumetric flow rate from PED extruder	—	Derived
Q_D	Drag volumetric flow rate	—	Derived
Q_N	Die restriction volumetric flow rate	—	Derived
Q_P	Poiseuille volumetric flow rate	—	Derived
p	Pressure in PED extruder	—	Derived
T_1	PED extruder barrel temperature, inlet	$65 \leq T_1 \leq 90^\circ\text{C}$	Process
T_2	PED extruder barrel temperature, outlet	$65 \leq T_2 \leq 90^\circ\text{C}$	Process
t	Time	—	—
v	PED deposition head speed	$0.1 \leq v \leq 10.0 \text{ mm/s}$	Process
W	PED extruder screw channel width	17.0 mm	Geometric
z	Height between nozzle and substrate	$0.1 \leq z \leq 2.0 \text{ mm}$	Process
Φ	PED material parameters	—	Material
X	PED geometric parameters	—	Geometric
Ψ	PED design variables	—	Design
Ω	PED process parameters	—	Process
α	Drag flow coefficient	$\pi D H W \cos \theta$	Derived
β	Pressure flow coefficient	$W H^3 / 12 L$	Derived
γ	Strain rate	—	Derived
ε	PED extruder die characteristic	—	Geometric
θ	Screw helix angle	18.5 deg	Geometric
μ	Material viscosity	$K T^n$	Material
ρ	Material density	1.145 g/cm ³	Material
ω	Die restriction coefficient	$\pi / 128 l$	Derived

characteristic coefficient, ε , and screw helix angle, θ , Eq. (3). Figure 4 shows the PED material delivery system's geometric parameters annotated and modeled as an unrolled helical screw

$$X = [d \ D \ H \ L \ l \ W \ \varepsilon \ \theta] \quad (3)$$

The 3D PCL pattern features specific macroscale (10^{-3} to 10^{-1} m) placement along the tool path and microscale (10^{-4} to 10^{-3}

m) cross-sectional width/height of the extruded filament. The macroscale pathway of material deposition is controlled by the motion system to be repeatable to less than 10 μm . Microscale design variables to characterize the printed filament's cross section are controlled by system's process parameters. The set of design variables geometrically define the printed filament's cross-sectional width, a , and height, b (Eq. (4))

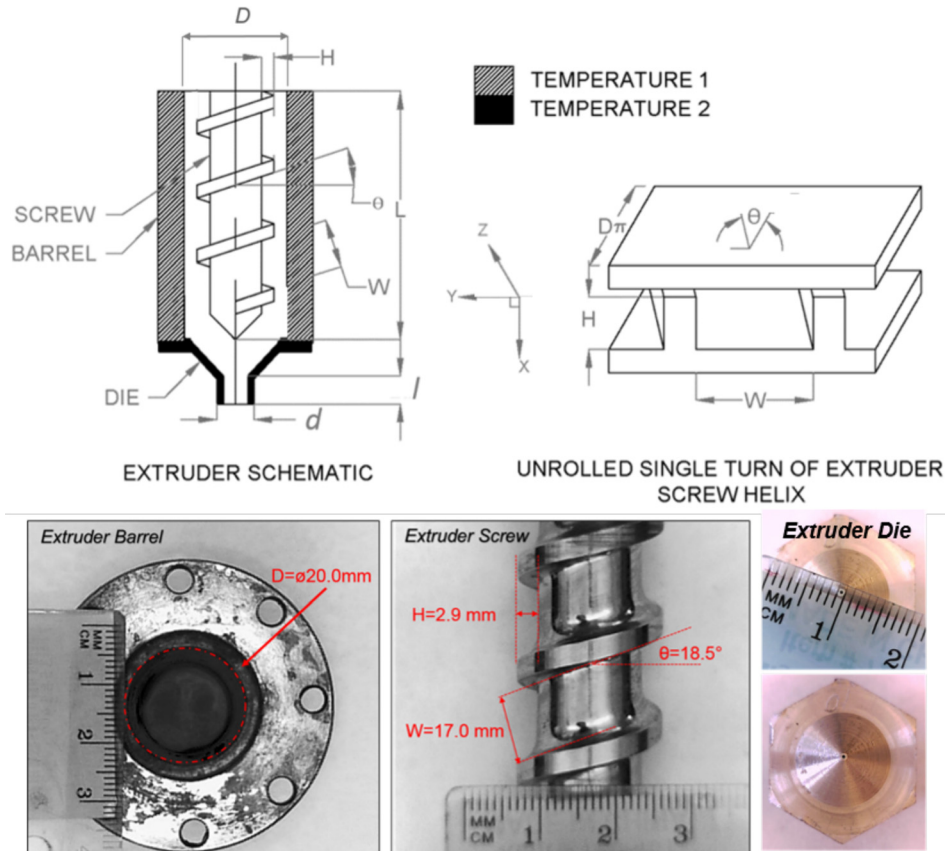


Fig. 4 PED material delivery system's extruder screw with geometric parameters annotated and modeled as an unrolled helical screw. Photographs present the PED extruder barrel, screw, and die with measurements.

$$\Psi(t) = [a(t) \quad b(t)] \quad (4)$$

The conservation of mass identifies the causality between process, geometric, material properties, and design variables. PCL flow from the PED extruder is equal to the mass of PCL accumulating as a filament on the substrate (Eq. (5))

$$\left(\frac{dm}{dt}\right)_{\text{from extruder}} = \left(\frac{dm}{dt}\right)_{\text{substrate accumulation}} \quad (5)$$

Assuming the PCL is a constant density fluid within PED operating conditions, the mass and volume flow rates are proportional. Therefore, net volumetric flow rate from the extruder, Q , is equal to the cross-sectional area of the printed filament modeled as half an ellipse is characterized by a major axis (width), a , and minor axis (height), b (Eq. (6))

$$Q = \frac{\pi abv}{4} \quad (6)$$

The design variable for height of the printed filament is directly controlled by the process parameter for the height of the PED extruder die outlet above the printing substrate (Eq. (7))

$$b = z \quad (7)$$

Flow through the PED extruder barrel is modeled as no-slip laminar flow through an unrolled screw helix with rectangular cross section. The restrictor die fixed to the barrel's outlet causes pressure, which resists the drag flow caused by the rotating screw,

defined as a Drag-Poiseuille flow. Further, the net volumetric flow rate through the extruder die is a function of the geometry of the restrictor die itself. Equation (8) presents the net volumetric flow on the printing substrate as a combination of the Drag flow, Q_D , and Poiseuille flow, Q_P , and equal to the flow through the restrictor die, Q_d

$$Q = Q_D - Q_P = Q_d \quad (8)$$

The Drag flow is the down channel component of the velocity integrated over the cross-sectional area. The Drag flow is a function of geometric parameters extruder barrel diameter, D , and screw pitch, θ , and process parameter screw speed, N . The geometric parameters are represented as a single drag flow coefficient, α (Eq. (9))

$$Q_D = \frac{\pi}{2} D H N W \cos \theta = \alpha N \quad (9)$$

Poiseuille flow is produced by a pressure gradient caused by flow restriction due to an extruder die, Eq. (10). Geometric parameters are represented as a single Poiseuille flow coefficient, β

$$Q_P = \frac{dV}{dt} = \frac{W H^3 p}{12 \mu_1 L} = \frac{\beta}{\mu_1} p \quad (10)$$

Flow due to restriction die is a function of viscosity, μ , pressure, p , and die geometry coefficient, ϵ , which is a function of diameter, d , and length, l , for a capillary dye, defines the flow rate through the extruder die. Geometric parameters are combined by the die flow coefficient, ω (Eq. (11))

$$Q_d = \frac{\varepsilon}{\mu_2} p = \frac{\pi d^4}{128 l \mu_2} p = \frac{\omega d^4}{\mu_2} p \quad (11)$$

2.2.2 Working Material Rheology. Polymer rheology is empirically defined from a parametric study of the temperature- and strain rate-dependence of viscosity within the PED's working conditions. The viscosity of liquid phase PCL is experimentally evaluated for thermal and mechanical sensitivity. Equation (12) defines the set of material properties relevant to PED as viscosity, μ . Viscosity is generalized as a function of strain rate, γ , and temperature, T

$$\Phi = [\mu = f(\gamma, T)] \quad (12)$$

PCL viscosity is measured using shear rate and temperature sweep to identify the dependence of each variable. The effect of a shear rate sweep using a Peltier plate setup classifies the material as Newtonian or non-Newtonian within the PED operating conditions.

PCL ($C_{26}H_{52}O_2$) pellets (Sigma-Aldrich) are loaded between the parallel plates of TA Instrument's AR2000ex Advanced Rheometer System. The controlled shear rate testing mode continuously deforms the fluid using a shear rate sweep to exponentially increase the rotation of the plate from 0.1 to 100 1/s over a 45-min-test period. The rheometer controls the rate of rotation (shear rate) and measures the motor's torque (shear stress) and temperature. The 65°C test condition is evaluated first. Temperature is increased by 5°C for each trial up to the maximum 90°C condition.

2.2.3 Profilometer Imaging. Optical surface interferometer Zygo NewView 6000 characterizes surface profiles, topology, and critical dimensions of the cross section of the PED system printed PCL pattern. Printed PCL patterns are placed on a glass slide and sputter coated with 5 nm thick layer of platinum-palladium prior to imaging.

2.3 Cell Culture, Encapsulation, and Imaging

2.3.1 Cell Culture and Fluorescent Labeling. Human hepatocytes of the HepG2 cell line (ATCC) are cultured in Eagle's Minimum Essential Medium base medium (ATCC) supplemented with

10% (v/v) fetal bovine serum (ATCC). Fluorescence based Live/Dead Viability/Cytotoxicity Kit for mammalian cells (Molecular Probes) qualitatively demonstrates cell viability. Two-color discrimination by the fluorescent probes ethidium calcein acetoxymethyl ester and ethidium homodimer-1 labels live cells as green and dead cells as red. Samples were analyzed using a Leica direct magnification inverted microscope (Leica) with UV source. Images of all cells were captured electronically by Insight 4.0Mp Monochrome digital camera (Spot Insight) and manufacturer provided imaging software.

2.3.2 Cell-Laden Medium Preparation. Cells are collected from culture dishes using Trypsin-ethylenediaminetetraacetic acid (EDTA) solution and centrifuged for 5 min at 1800 rpm. Cell re-suspended to a final concentration of 2.0×10^6 cells per mL fresh cell culture medium before injection into fully embedded microfluidic manifold.

2.3.3 Bioprinter Working Solution Preparation. Alginate matrix prepared from 6.0% (w/v) alginic acid sodium salt from brown algae in deionized water is sterilized by serial filtration through 0.2 μ m mesh (Van Water and Nat rogers, VWR) [54]. Cells are mixed with alginate for final concentration of 2.0×10^6 HepG2 cells per mL aqueous alginate solution. Cell-laden matrix is prepared and loaded into multinozzle deposition direct cell writing system and printed. After printing, channels are flooded with 5.0% (w/v) calcium chloride in deionized water cross-linking solution to polymerize cell-laden matrix.

3 Results

3.1 Empirical Characterization of PED Working Material Rheology. Figure 5 shows photographs of the AR2000ex Advanced Rheometer System with the PCL pellets loaded on the lower of two Peltier plates and the results of strain- and temperature-sensitivity of PCL through the functional PED operating windows. PCL viscosity within the PED operating windows significantly decreases as temperature increases and does not present significant strain-dependence. Each 5°C increase through the

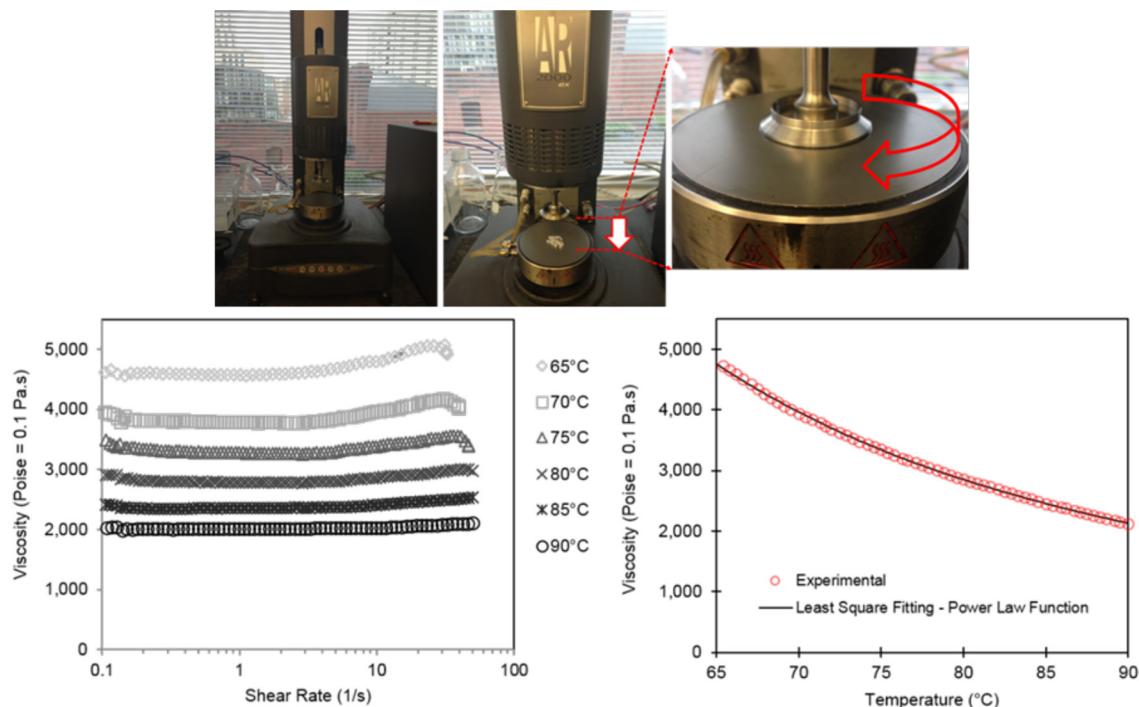


Fig. 5 (Top) AR2000ex Advanced Rheometer System photographs with PCL pellets loaded on the lower of two Peltier plates. (Bottom) Strain- and temperature-sensitivity of PCL through the functional PED operating windows.

65–90 °C temperature sweep decreases the PCL viscosity. The strain rate sweep from 0.1 1/s to the maximum fluid shear rate does not significant effect on viscosity. PCL in the PED operating conditions is classified as a Newtonian fluid with thermal sensitivity modeled using the power law with a coefficient, K , and index, n , to define the viscosity as a function of temperature T . Least square fitting regression using a power law model of the experimental data determined the coefficient and index (Eq. (13))

$$\mu = KT^n = 1.4 \times 10^8 T^{-2.46} \quad (13)$$

3.2 Fabrication Capability

3.2.1 Control of PED Extrusion Process. Equation (14) combines power law of viscosity as a function of temperature equation (13), the net volumetric flow rate equation (8), Drag flow equation (9), Poiseuille flow equation (10), and flow through the restrictor die equation (11) to define the net volumetric flow rate from the extruder to the printing substrate as a function of the system's parameters.

$$Q = \alpha N \left(1 - \frac{1}{d^4 \frac{\omega}{\beta} \left(\frac{T_1}{T_2} \right)^n + 1} \right) \quad (14)$$

The linear sensitivity of screw speed and flow rate presented in the derived model, Eq. (14), is validated by extrusion of PCL through 80 °C PED extruder barrel with a Ø350 µm restrictor die. Figure 6 shows the mass flow rate from the PED extruder as a function of the controllable PED process parameter screw rotation speed. The derived model and experimental PED extruder flow rates demonstrate strong agreement in trend and quantitative estimation, especially for screw speeds 10–120 rpm. Applications of

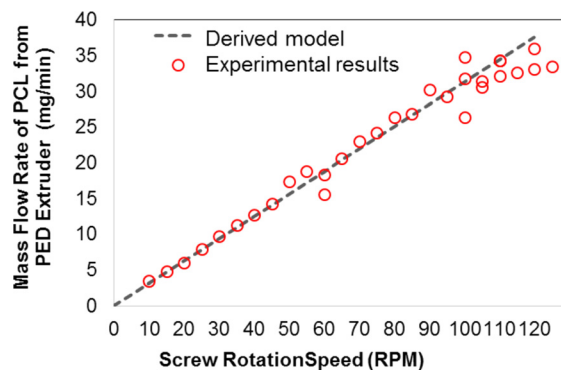


Fig. 6 Comparison of experimental and derived model (Eq. (14)) results for process control PED process parameter screw rotation speed's effect on the mass flow rate from the PED extruder, where both the heating elements are set to 80 °C and Ø350 µm restrictor die.

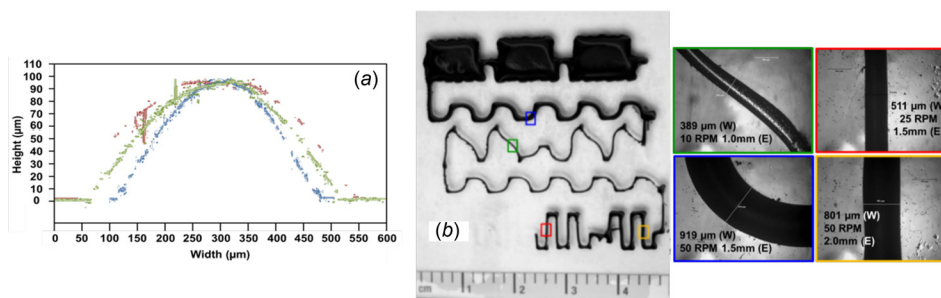


Fig. 7 (a) Profiles of three PED printed PCL filament with the same extruder height above the substrate and variable printhead speed. (b) Open channels in PDMS substrate fabricated by replica molding of PED printed PCL template. Channel Width (W), PED screw speed RPM, and height of the outlet tip above the substrate (E) annotated.

the derived model for process control using Eq. (14) for temperatures other than the surveyed 80 °C for both heating elements require additional validation.

3.2.2 Effect of PED Process Parameters. The set of PED process parameters necessary to define the printed PCL filament's cross section is independently adjustable and dynamically controllable over the course of the build cycle. For printed filament cross sections whose width is greater than or equal to height, the height of the extruder above the substrate and resulting printed PCL filament height are equal, Eq. (7). The width of the printed filament is a function of the flow rate from the extruder screw, printhead speed, and height. Therefore, the height and width can be independently adjusted during a single build cycle, if the width is greater than the height.

Figure 7 shows the measured profile of three PED printed PCL filament using variable process parameters. The width of the most narrow horizontal profile (fastest printhead speed-least specific area accumulation) is less than the widest horizontal profile (slowest printhead speed), and the intermittent horizontal profile width is generated by the intermittent printhead speed. The height of the printed filament is constant between all three samples. Figure 7 also presents photographs of the open channels in the PDMS substrate fabricated using the printed PCL template. The variability of open channel and flow pattern is enabled by the dynamic adjustment of PED screw speed during a single build cycle.

3.2.3 3D Manifold of Channels. Layer-by-layer the PED system builds internal features in a 3D scaffold, which becomes 3D internal architecture of an open network of channels after replica molding. Figure 8 shows the photograph of a PDMS substrate with a network of open channel templated by a three-layer PED printed PCL template. The template is a stack of three square wave patterns. The PDMS substrate is cross-sectioned as annotated and photographs of the sections are also presented. The top layer of the manifold is exposed. However, the deeper layers are not accessible for cell printing or plasma treatment techniques.

3.3 Flow Tested for Leakage. Perfusion of fluid through the fully constructed microfluidic device demonstrates the connectivity of the open network channels. The two types of construction—with (surface exposed channels) and without cover components (fully embedded in cured PDMS) presented in this research maintain a leak-proof seal during flow up to 10 µL/hr. The flow rate is sufficient to introduce fresh nutrients, remove waste, and maintain a residence time of several hours of soluble nutrients.

Fully embedded microfluidic manifolds—do not require an additional cover component—maintain a leak-proof seal up to 2.0 mL/min. The fully embedded microfluidic manifold and resultant increased maximum flow rate capability offer active physical loading by hydrodynamic stress testing and present the possibility of sub-1-min-residence times for soluble cues in the cell-laden device. Figure 9 presents a time lapse photo sequence of PDMS microfluidic device being perfused. Each frame contains

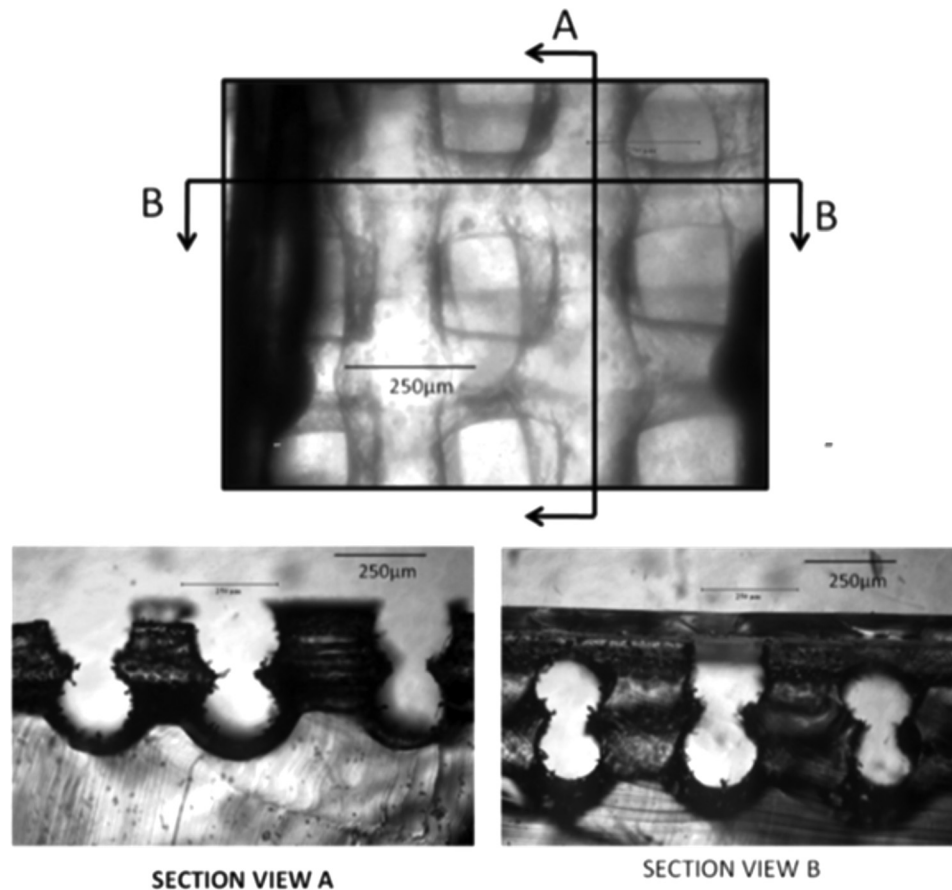


Fig. 8 PDMS substrate with embedded 3D manifold fabricated by replica molding of three-layer PED printed PCL template. Cross section of the PDMS substrate presents interconnected network of orthographic channels.

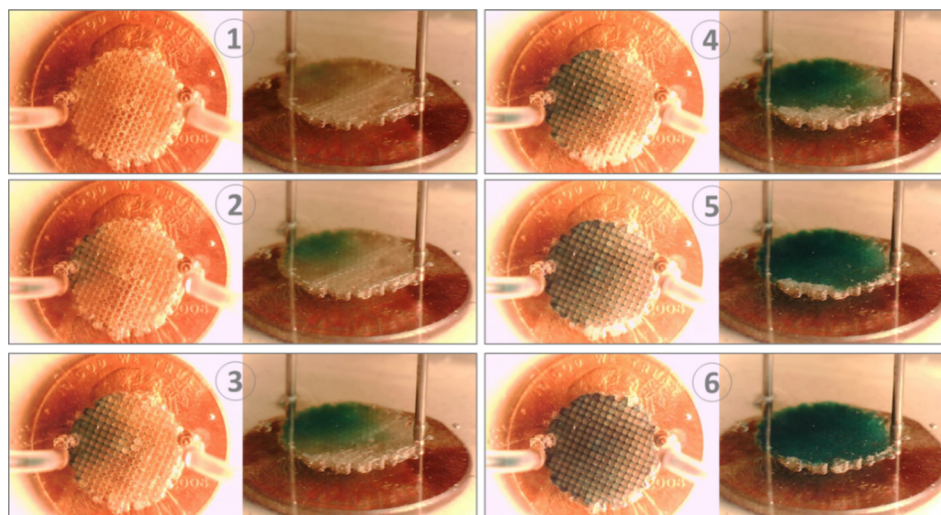


Fig. 9 Time-lapse photo sequence of PDMS microfluidic device being perfused. Each frame contains a top (left) and front (right) views of the device.

a top (left) and front (right) view of the device. The fully embedded microfluidic manifold and its significantly higher leak resistance are more.

3.4 Bulk Cell Injection after Fabrication

3.4.1 Final Construction. Cell-laden microfluidic device is fabricated by (1) the FDM technique PED to print a sacrificial

PCL template for PDMS replica molding, (2) insert a minimum of two capillary tips to perfuse and drain liquid medium through the open channel manifold embedded in the cured PDMS substrate, (3) plumb with silicone tubes to a syringe pump for controlled, dynamic perfusion, and (4) sterilize and inject cell-laden solution.

3.4.2 Image Cells Attached to Channel Walls. Hepatocyte-laden medium is injected into the device and left overnight for

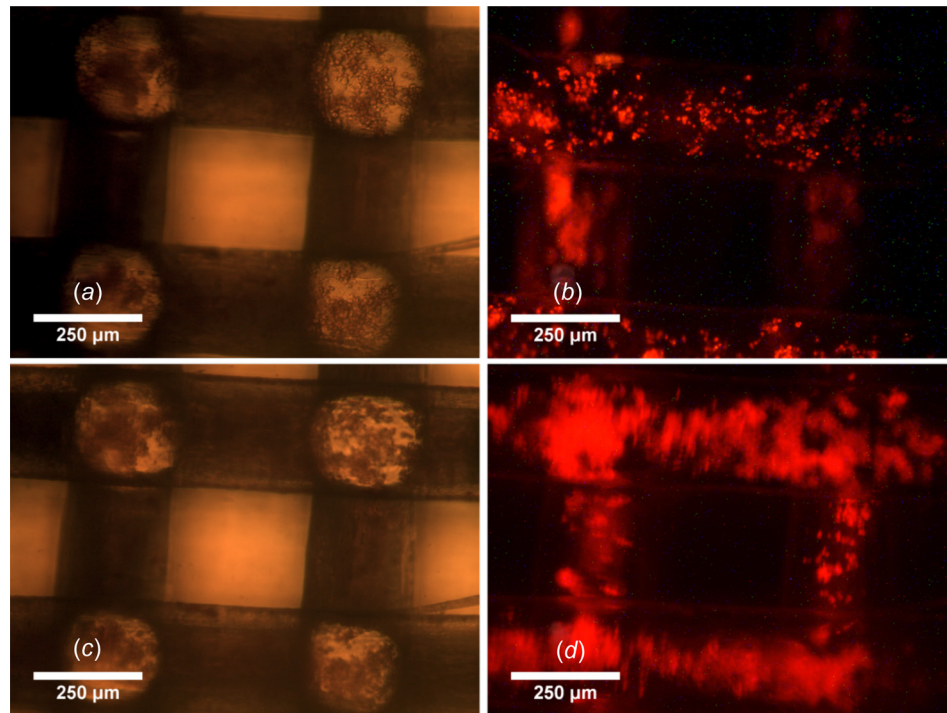


Fig. 10 3D PDMS manifold seeded by injection of cell-laden medium. All four images are the same microscope view field: ((a) and (c)) light and ((b) and (d)) fluorescent images on two focal planes ((a) and (b)) closer and ((c) and (d)) farther focal plane.

cell attachment in 37 °C and 5% CO₂ incubation. After 12 hrs, the device is perfused with fresh cell culture medium at a rate of 1.0 mL/hr for 3 hrs to remove nonadherent cells. After perfusion, the cells are fixed in 70% ethanol and fluorescently labeled using Live/Dead Viability/Cytotoxicity stain for mammalian cells (Molecular Probes). Figure 10 shows light and fluorescent images of a single view field focused on two different focal planes to demonstrate cells residing on the channel walls through the 3D depth of the manifold.

3.5 Controlled Cell Seeding During Fabrication

3.5.1 Final Construction With Bioprinting. The bioprinting technique leverages the open channel as a mold to hold extruded material to preserve the shape and mechanical rigidity of the printed product. The extrusion process requires the channels be surface exposed, not fully embedded. Open channels are fabricated using the PED/replica molding process as previously described. After the PDMS substrate is constructed, plasma surface treatment oxidizes the PDMS substrate methyl groups to form silanol groups to improve sealing and wettability [55]. PDMS substrate is placed in RF plasma chamber (Harrick Plasma) for 90 s prior to printing.

Cell printing by multinozzle deposition produces explicitly designed 3D patterns of cell-laden matrix in microfluidic device [56,57]. The material delivery system is mounted to a high precision positioning system to print the cells into microfluidic channels. The design input model for cell printing tool path and the PED tool path used to fabricate the PCL pattern and template the open network of channels are the same.

After printing, the nanopore assemblies (Upchurch Scientific) mounted to a PDMS cover are manually aligned to the microfluidic network's inlet/outlet and fastened together in a polycarbonate brace to form a microfluidic circuit. Nanopores are plumbed to a syringe housed in a programmable syringe pump for controlled diffusion.

3.5.2 Imaging Printed Cells. Microscopy of the cells within the device after printing demonstrates the success of multinozzle cell printing to assemble viable cells in controlled arrangements in the channel cross section and through the channel network length from inlet to outlet. For heterogeneous bioprinting, a suspension of viable cells is purposely lysed to visually discriminate the extrusion of two separate bioprinter nozzles during imaging. Hepatocyte-laden alginate is prepared as previously described. However, before multinozzle printing the cell-laden solution is split into two equal volumes. One volume is stored in 37 °C water bath. The other volume is repeatedly frozen/thawed between −210 °C liquid nitrogen and 37 °C water bath to lyse cells. Each solution (healthy and lysed) is loaded into separate nozzles and printed into alternating channels of a square wave network. Cell position identified by Live/Dead staining post-printing.

Figure 11 shows light and fluorescent images of the fabricated channels and bioprinted cells for both homogeneous and heterogeneous arrangements. Fluorescent imaging confirmed that viable hepatocytes were printed into microfluidic channels using single nozzle, homogeneous cell printing. No necrotic subpopulations were qualitatively observed in homogeneous arrangement. The fluorescent probes successfully demonstrate that the two distinct cell populations were arranged in series through the channel network length.

4 Limitations

The presented research shows the PED as a viable method to control internal features of a 3D construct and—after replica molding—the internal features of an open network of channels. The minimum resolution of the PED process is orders of magnitude greater than other microfabrication techniques, such as photolithography. However, the FDM techniques such as the PED system offer greater accessibility, biologically relevant resolution, lower expense, do not require wet lab space, toxic chemicals, mask aligners, or radiation sources. PED is not the only, but also a

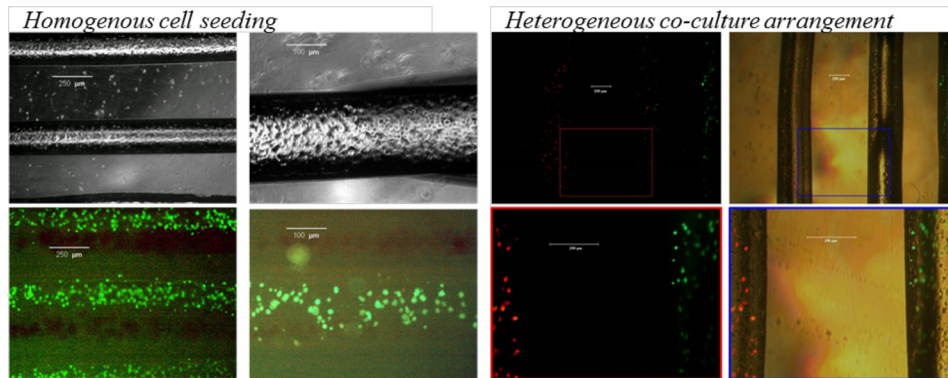


Fig. 11 Cells bioprinted in open channels labeled with fluorescent probes to demonstrate controlled cellular arrangement in the open channel network using single and dual nozzle printing

successful and accessible technique to produce biologically relevant features (10^{-4} to 10^{-3} m) and after replica molding—microfluidic devices.

The principle limitation of the proposed results is the inability to integrate the processes to bioprint cells during fabrication and the construction of 3D architecture to guide cells after fabrication. The bioprinting process to dispense cells into the channels requires the channels be exposed to the surface—and reachable by the bioprinter's rigid dispensing tip.

Cell printing controls the geometric pattern and specific area of cell-laden matrix in channels with greater resolution and specificity than the alternative bulk cell suspension injection for adhesion to channel walls over several hours. To control cell arrangement from inlet to outlet of multiple phenotypes, a multinozzle deposition bioprinting system automates 3D deposition relative to global coordinate space and the other working solution being printed.

5 Conclusion

The combined PED and replica molding process defined in the presented work produces an open network of channels embedded in a transparent elastomer substrate. The channel network is templated by a sacrificial 3D printed scaffold, whose architecture includes specific macroscale (10^{-3} to 10^{-1} m) placement along a tool path and microscale (10^{-4} to 10^{-3} m) cross-sectional width/height of the extruded filament. The feasible set of design parameters are defined from the PED's hardware and operating windows. The macroscale pathway of material deposition is controlled by the motion system to be repeatable to less than $10\text{ }\mu\text{m}$. PCL's rheological sensitivity to temperature and strain rate within the PED's process windows is evaluated, and empirically defined coefficients to relate process parameters and viscosity are defined. Geometric parameters are measured from the as-built system. A derived model of the design, process, material, and geometric parameters is presented and compared to experimentally measured extrusion rates. Microscale design variables to characterize the printed filament's cross section are controlled by system's process parameters. The dependence of the width and height of the printed filament is diminished, as each parameter is independently controllable and automated to be varied over the build cycle, and consequently, throughout the construct. Investigation of the cross-sectional profile of a printed filament shows the height of the structure can be maintained constant, while decreasing the width using the process parameters. After the printed pattern is replica molded, the pattern is thermally and chemically removed leaving an open network of channels fully embedded in the elastomer. A fully embedded network does not require an additional cover component or alignment of a cover component. The simplified construction improves the leak resistance of the device. The substrate

and programmable syringe pump collectively form a microfluidic system; which is perfused to 2.0 mL/min without leakage. Cells adhere to the channels walls throughout the internal features of the network. The presented process expands the set of design variables to include internal features and cross-sectional profile in a single step of replica molding. Surface exposed channels serve as a mold for bioprinted cell-laden matrix. Multinozzle deposition bioprinting is a successful technique to arrange multiple cell types in preprogrammed patterns between the inlet and outlet of a device for heterocellular culture.

Nomenclature

a	= PED printed filament cross section width
b	= PED printed filament cross section height
d	= PED restrictor die diameter
D	= PED extruder barrel diameter
H	= PED extruder screw channel depth
l	= PED restrictor die length
L	= PED extruder pumping zone length
m	= mass
N	= PED extruder screw speed
p	= pressure
Q	= volumetric flow rate
Q_d	= volumetric flow rate through restrictor die
Q_D	= drag flow rate
Q_P	= Poiseuille flow rate
t	= time
T_i	= PED temperature of material delivery system
v	= PED extruder head speed
W	= PED extruder screw channel width
X	= the set of PED extruder geometric parameters
z	= PED restrictor die outlet above printing substrate
α	= drag flow coefficient
β	= Poiseuille flow coefficient
γ	= strain rate
ε	= PED restrictor die characteristic coefficient
θ	= PED extruder screw helix angle
μ	= viscosity
Φ	= the set of PED working material parameters
Ψ	= the set of PED design variables
Ω	= the set of PED process parameters
ω	= restrictor die flow coefficient

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